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Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS

Half Past Paper – Week 17

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1a	Simplifying expressions using index laws	1	U662
1b	Simplifying expressions using index laws	2	U662
1c	Solve single inequalities	2	U759
3	Finding error intervals	2	U657
5a	Tree diagrams for independent events	2	U558
5b	Tree diagrams for independent events	2	U558
7	Calculating the mean	3	U291
9	Finding unknown sides in right-angled triangles, Finding the volume of prisms	5	U283, U174
11a	Combining transformations	3	U766
11b	Combining transformations	1	U766
13a	Drawing graphs of inequalities	4	U747
13b	Drawing graphs of inequalities	1	U747
15	Converting recurring decimals to fractions	2	U689
17	Calculating averages from histograms	4	U267
19	The area rule, Finding the perimeter and area of similar shapes	4	U592, U630
21a	Calculating with ratios and algebra	2	U676
21b	Factorising quadratic expressions, Calculating with ratios and algebra	3	U858, U676
	Total	43	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 (*a*) Simplify $n^3 \times n^5$

.....

(1)

(*b*) Simplify $\dfrac{c^3 d^4}{c^2 d}$

.....

(2)

(*c*) Solve $\dfrac{5x}{2} > 7$

.....

(2)

(Total for Question 1 is 5 marks)

3 A number, m, is rounded to 1 decimal place.
The result is 9.4

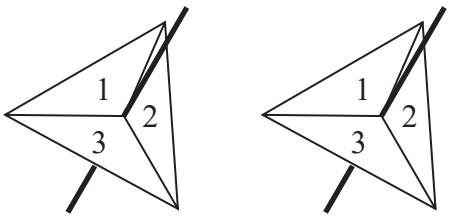
Complete the error interval for m.

..... $\leqslant m <$ .....

(Total for Question 3 is 2 marks)

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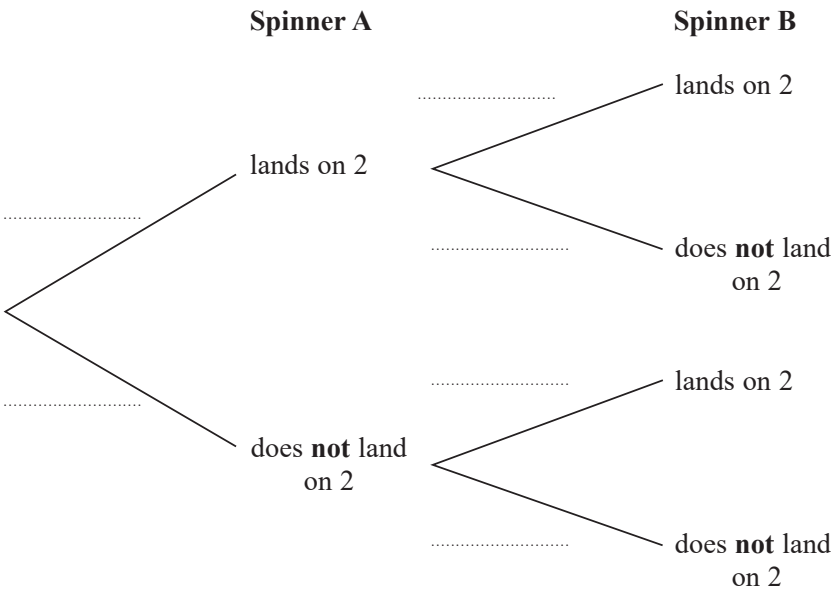
5 Amanda has two fair 3-sided spinners.



Spinner A **Spinner B**

Amanda spins each spinner once.

(*a*) Complete the probability tree diagram.



(2)

(*b*) Work out the probability that Spinner A lands on 2 and Spinner B does **not** land on 2

.....

(2)

(Total for Question 5 is 4 marks)

7 There is a total of 45 boys and girls in a choir.

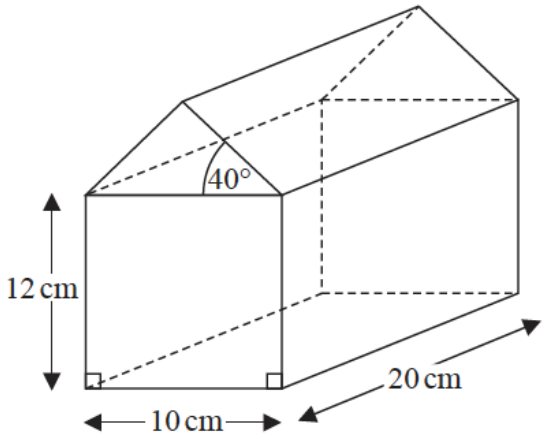
The mean age of the 18 boys is 16.2 years.
The mean age of the 27 girls is 16.7 years.

Calculate the mean age of all 45 boys and girls.

..... years

(Total for Question 7 is 3 marks)

9 The diagram shows a prism.



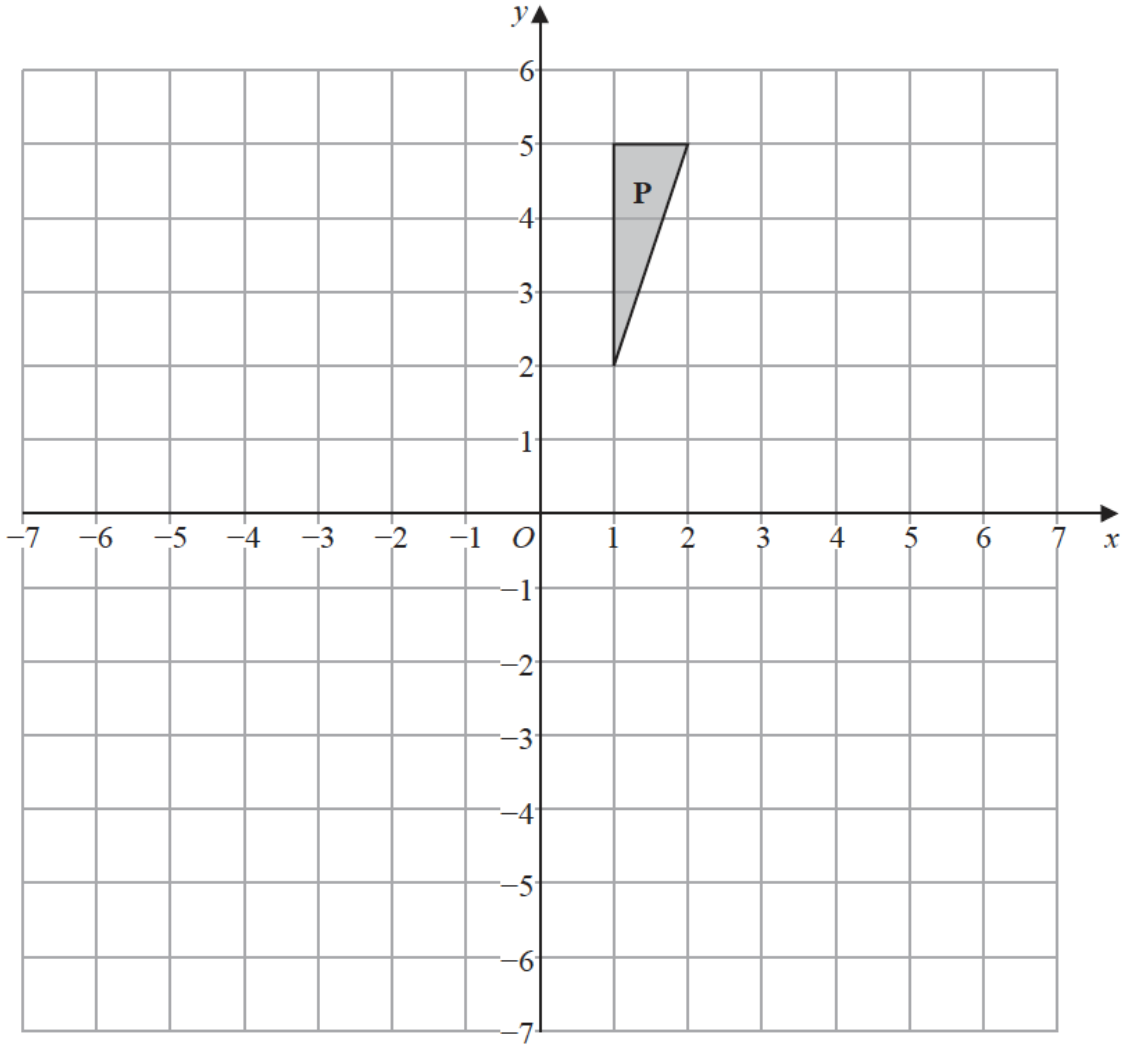
The cross section of the prism has exactly one line of symmetry.

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

..... cm^3

(Total for Question 9 is 5 marks)

11 The diagram shows a triangle **P** on a grid.



Triangle **P** is rotated 180° about (0, 0) to give triangle **Q**.

Triangle **Q** is translated by $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$ to give triangle **R**.

(*a*) Describe fully the single transformation that maps triangle **P** onto triangle **R**.

.....

.....

(3)

Under the transformation that maps triangle **P** onto triangle **R**, the point *A* is invariant.

(*b*) Write down the coordinates of point *A*.

(..... , ...)

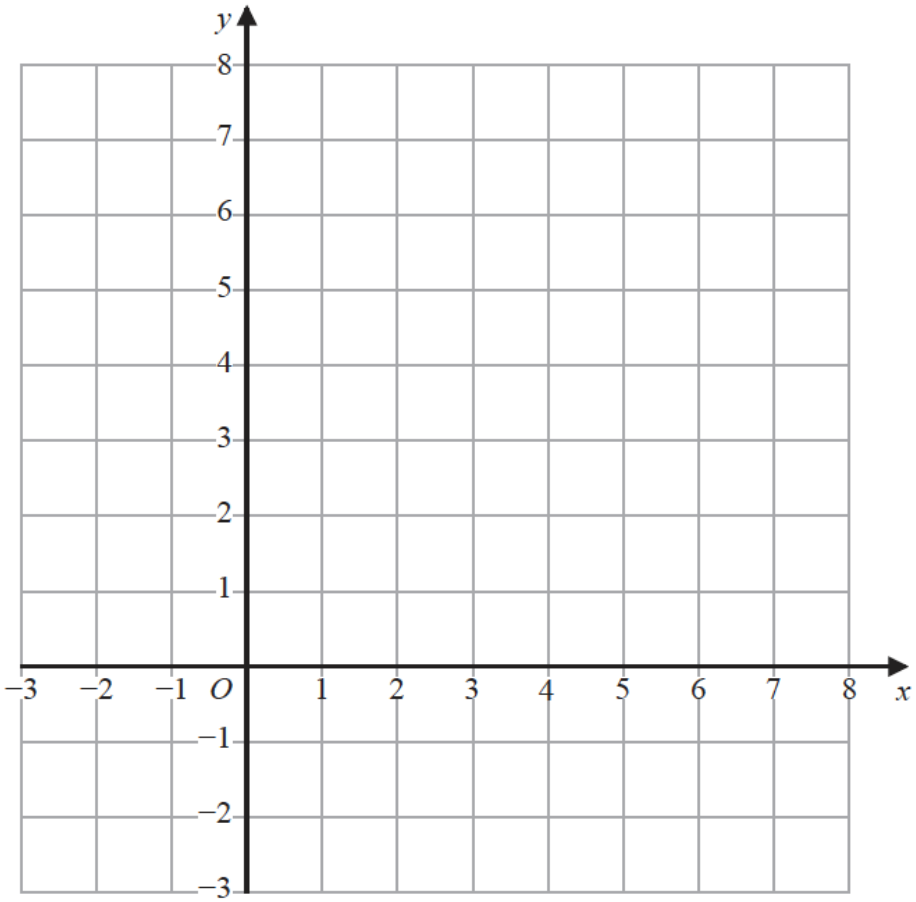
(1)

(Total for Question 11 is 4 marks)

13 (*a*) On the grid show, by shading, the region that satisfies all these inequalities.

$x \geqslant 0$ $x \leqslant 2$ $y \leqslant x + 3$ $2x + 3y \geqslant 6$

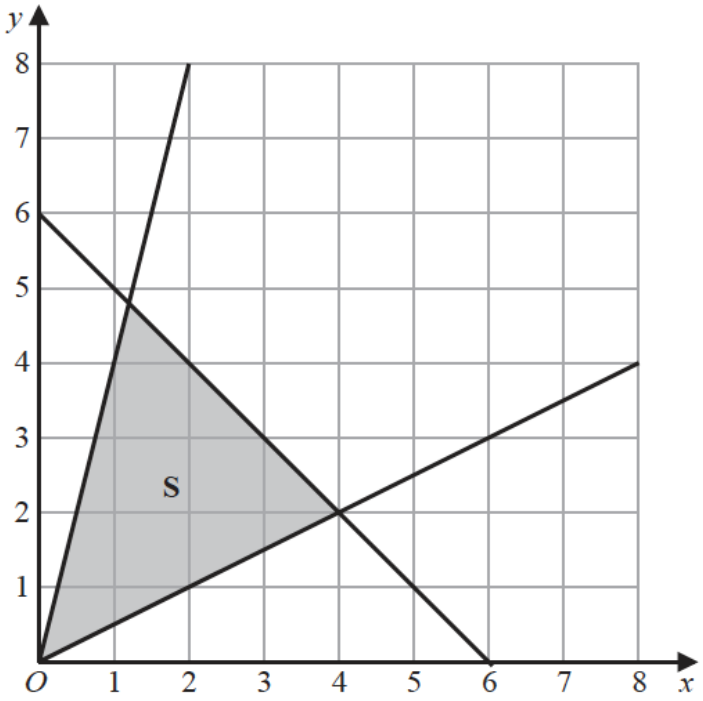
Label the region **R**.



(4)

(*b*) The diagram below shows the region **S** that satisfies the inequalities

$$y \leqslant 4x \qquad y \geqslant \frac{1}{2}x \qquad x + y \leqslant 6$$



Geoffrey says that the point with coordinates (2, 4) does not satisfy all the inequalities because it does not lie in the shaded region.

Is Geoffrey correct?
You must give a reason for your answer.

.....

.....

.....

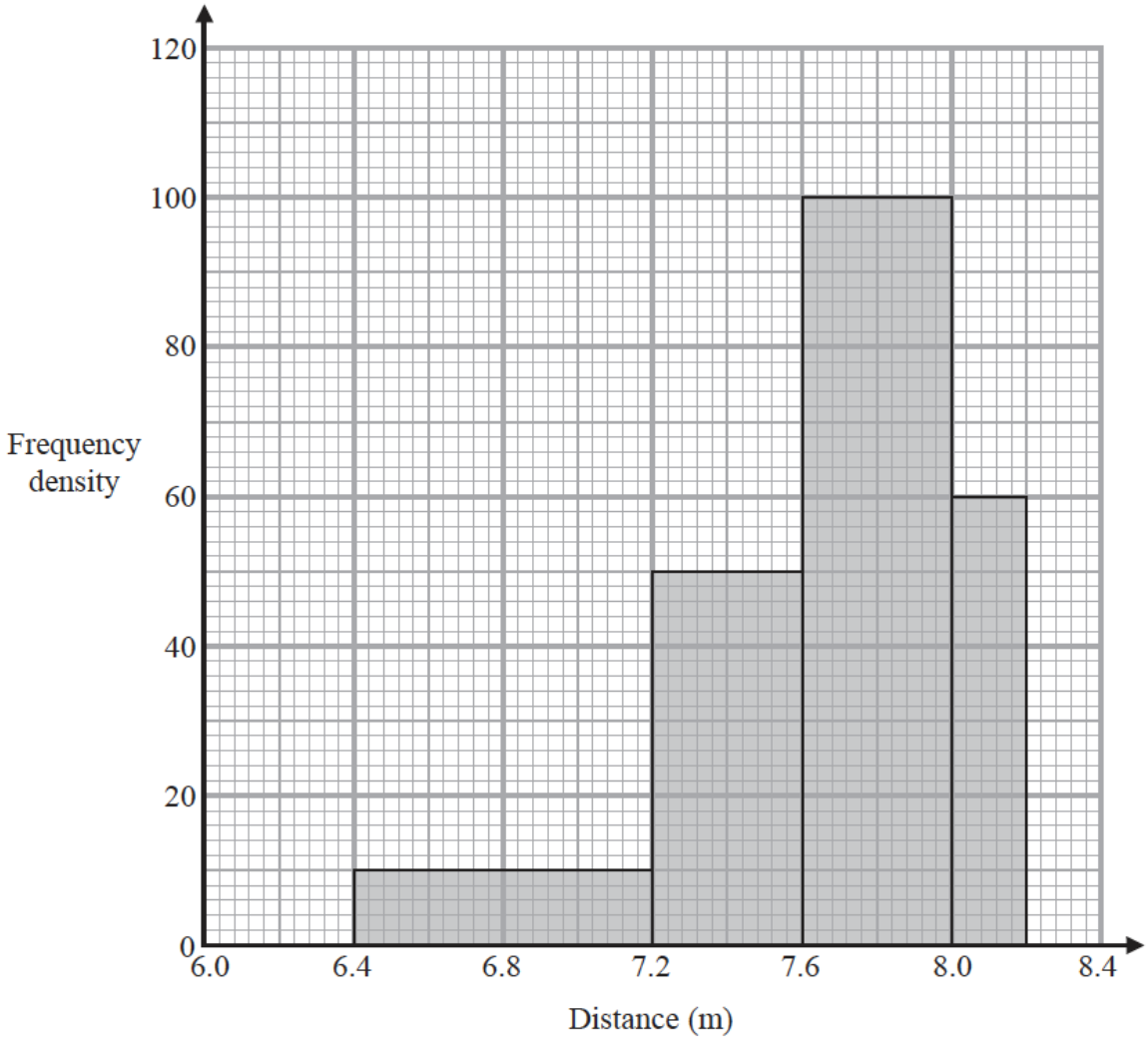
(1)

(Total for Question 13 is 5 marks)

15 Prove algebraically that $0.7\dot{3}$ can be written as $\frac{11}{15}$

(Total for Question 15 is 2 marks)

17 The histogram gives information about the distances 80 competitors jumped in a long jump competition.

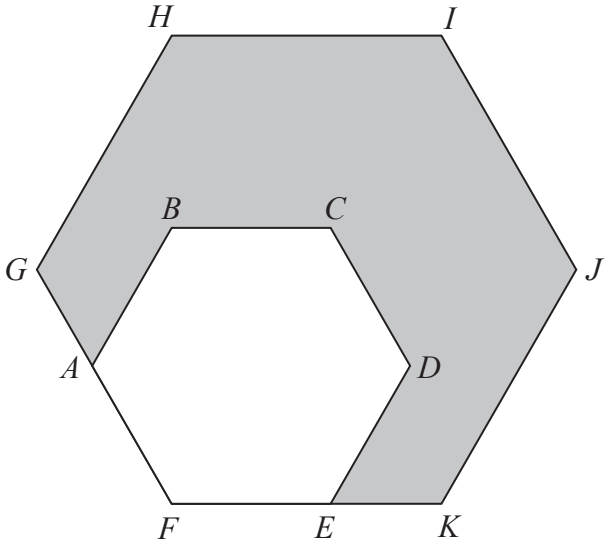


Calculate an estimate for the mean distance.

..... m

(Total for Question 17 is 4 marks)

19



ABCDEF is a regular hexagon with sides of length x.
This hexagon is enlarged, centre F, by scale factor p to give hexagon *FGHIJK*.

Show that the area of the shaded region in the diagram is given by $\frac{3\sqrt{3}}{2}(p^2-1)x^2$

(Total for Question 19 is 4 marks)

21 $5c + d = c + 4d$

(*a*) Find the ratio $c : d$

.....

(2)

$6x^2 = 7xy + 20y^2$ where $x > 0$ and $y > 0$

(*b*) Find the ratio $x : y$

.....

(3)

(Total for Question 21 is 5 marks)

TOTAL FOR PAPER IS 43 MARKS